

Bachelor's Thesis

Dispersion Compensation in IM/DD Systems Using the Gerchberg-Saxton Algorithm

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Ich versichere hiermit wahrheitsgemäß, die Arbeit bis auf die dem Aufgabensteller bereits bekannte Hilfe selbständig angefertigt, alle benutzten Hilfsmittel vollständig und genau angegeben und alles kenntlich gemacht zu haben, was aus Arbeiten anderer unverändert oder mit Abänderung entnommen wurde.

München, 23.02.2026

Ort, Datum

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(Ivan Lomakov)

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Abstract

Intensity Modulation/Direct Detection systems (IM/DD) are a type of optical communication systems, that are critical for high-capacity, short-reach optical transmissions. The physical properties of such systems subject them to various disruptive effects. These include chromatic dispersion - a linear effect that introduces frequency-dependent phase shifts that cause pulse broadening, thus distorting the signal. Such distorted signal is usually characterized by malformed impulses that negatively affecting the system's capability to accurately transmit data.

This paper investigates a method that utilizes the already well-documented Gerchberg-Saxton-Algorithm in order to mitigate chromatic dispersion. This algorithm, originally developed for the purpose of phase retrieval, can be adapted to exploit the deterministic nature of chromatic dispersion in order to pre-distort the transmitted signal such that it is transformed into its intended shape at the receiver. In order to assess its usefulness, we develop a simulation of an optical communication system and iteratively apply the said algorithm.

We evaluate the efficacy of this approach for mitigating the chromatic dispersion-induced distortion in order to assess it as a potentially promising alternative to existing methods. We also go over its limitations, as well as the possible areas of improvement.

Keywords: chromatic dispersion, IM/DD systems, Gerchberg-Saxton algorithm, transmitter predistortion, optical fiber communications, intersymbol interference

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1 Introduction

Optical communication systems use electromagnetic waves, i.e light, to propagate information through different forms of medium. The rapid adoption of the Internet and personal computers during the 1990's and early 2000's created a previously unseen demand for transfer of large batches of data at extremely high speeds, driving further research and advancements in optical fiber communication technology.

Modern optical communication is used in many different domains ranging from telecommunications to internet and data center interconnects. At their core these systems comprise three fundamental components: a transmitter that encodes the message into an optical signal, a channel that serves as a propagation medium, and a receiver that decodes the optical signal back into electrical form for further processing. Numerous implementations have been engineered over the years in order to address different constraints. These can range from the channel length and materials being used to requiring a specific data rate.

The following thesis focuses on Intensity Modulation/Direct Detection (IM/DD) systems, which have been developed for short and medium range communication (typically over a few tens of kilometers) over an optic fiber and are frequently used due to their cost-efficiency and compactness. In IM/DD systems, the transmitter modulates the optical intensity (amplitude) that varies depending on the specific symbol being transmitted. The system's channel is usually made up of standard single-mode fiber, which eliminates the possibility of otherwise common types of signal distortion (e.g modal dispersion), making it suitable for our simulation. The receiver is comprised of a photodiode (*square-law-detector*), which is used to convert the incoming optical signal into the electric domain.

The performance of optical communication systems is fundamentally limited by distortions that occur during the transfer of information in the channel. These arise primarily from the physical properties of optical fiber and system components, with their severity depending on factors such as transmission distance and data rate. Among them is chromatic dispersion (CD) - a phenomenon that always occurs in optical fibers and causes frequency dependent propagation delay. This means that various frequency components of the signal arrive with different delay, introducing phase distortion that manifests itself in malformed waveforms and transmission errors .

Various CD compensation techniques have been developed, operating either at the receiver to compensate dispersion effects or at the transmitter through predistorting the signal. Building off of the second option, we analyze the Gerchberg-Saxton (GS) algorithm adopted for CD compensation. The traditional GS algorithm [GS72] is commonly used to address a different problem within communication systems called *phase retrieval*¹.

This thesis presents the implementation of the GS Algorithm and evaluates its performance within the simulated IM/DD system. We start off by introducing the necessary theoretical concepts in Chapter 2, before delving into the specifics of the algorithm's implementation in Chapter 3. Chapter 4 demonstrates the obtained results and explains the significance for the initial problem setting. Chapter 5 addresses some of the potential improvements to our methodology, as well as touching upon existing literature that describes similar solutions, before moving onto Chapter 6, where we discuss the obtained results and draw final conclusions regarding the usefulness of our solution.

¹Phase retrieval is the problem of reconstructing a signal's phase solely from its amplitude measurement in a way that fits a set of constraints. It is essential in the field of optics, imaging, and diffraction where the phase information of the signal is lost for various reasons.

2 Theoretical Background

The following chapter breaks down the theoretical concepts behind the numerical simulation used in this work. We go over the mathematical model of an optical channel, as well as the background of chromatic dispersion and its underlying physical properties. Lastly, we present the fundamental principles of the GS algorithm, followed by a modified implementation tailored to chromatic dispersion compensation.

2.1 Optical communication systems

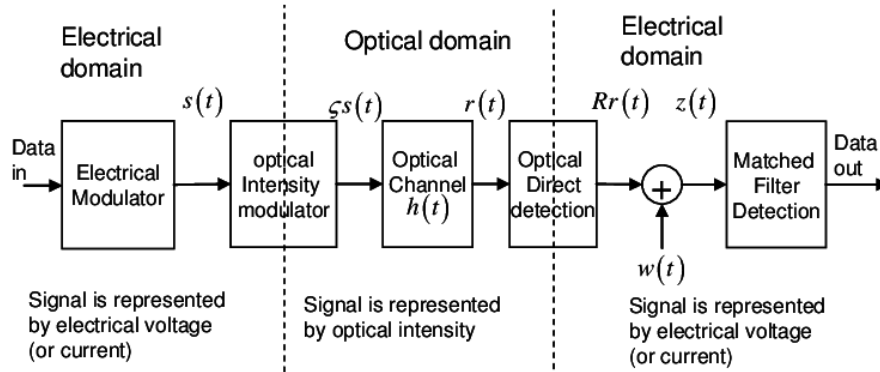


Figure 2.1: Example of an optical comms system[GLS⁺16].

2.1.1 Channel Definition

In order to fully understand the mathematical model of our simulation, we first need to define the channel. It serves as a medium for the signal to propagate through, while also being responsible for disruptive effects that occur during the propagation. The general assumption to be made, is that if the channel $h(t)$ is linear, then its interaction with the signal $x(t)$ can be modeled via convolution, defined as follows:

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (2.1)$$

Since the signal we are dealing with is both *finite* and *causal*, we can redefine the integral bounds as $[0, k]$ with k being the duration of the signal.

Recall that when dealing with signals, we often have to deal with their representations in both frequency and time domain. Transitioning between the domains requires the application of the *Fourier Transform* and the *Inverse Fourier Transform* respectively.

$$X(\omega) = \mathcal{F}\{x(t)\} = \int_{-\infty}^{\infty} x(t) e^{-i\omega t} dt \quad (2.2)$$

$$x(t) = \mathcal{F}^{-1}\{X(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{i\omega t} d\omega \quad (2.3)$$

It is worth noting that convolution can also be performed in frequency domain via multiplication, yielding better runtime complexity when compared to time-domain convolution ($O(n^2)$ vs $O(n \cdot \log(n))$). This allows us to define the transfer function of our channel as $H(f)$ and our signal as $X(f)$. Thus the interaction of a signal with the corresponding channel can be defined as follows.

$$Y(f) = X(f) H(f) \quad (2.4)$$

The transfer function $H(f)$ is an extremely important and versatile component of our system, as it allows us to define channel responses that differ by how they respond to different frequencies f . This is crucial for being able to model different linear effects.

2.1.2 Chromatic Dispersion

Chromatic dispersion (CD) is a fundamental impairment in optical fiber transmission that arises from the wavelength-dependent refractive index of silica fiber. This effect causes different spectral components of an optical pulse to propagate at different group velocities, resulting in temporal pulse broadening as the signal travels along the fiber (see Figure 2.2).

Chromatic dispersion (CD) is a linear effect, satisfying the superposition principle as described in Equation 2.5. This thesis considers a short-reach link and operating conditions under which other non-linear propagation effects are negligible. Hence, the channel is characterized entirely through its frequency-domain transfer function, which describes how different spectral components accumulate phase shifts proportional to their frequency. This linear model is valid for shorter distance communication systems.

$$H\{\alpha x_1(t) + \beta x_2(t)\} = \alpha H\{x_1(t)\} + \beta H\{x_2(t)\} \quad (2.5)$$

The propagation of a frequency dependent phase delay through a dispersive fiber can be described by the propagation constant $\beta(\omega)$, which varies with angular frequency ω . Expanding $\beta(\omega)$ in a Taylor series around the carrier frequency ω_0 yields:

$$\beta(\omega) = \beta_0 + \beta_1(\omega - \omega_0) + \frac{\beta_2}{2}(\omega - \omega_0)^2 + \frac{\beta_3}{6}(\omega - \omega_0)^3 + \dots \quad (2.6)$$

For shorter transmission distances and data rates considered in this thesis, the second-order term β_2 dominates the dispersion behavior. This means that higher-order terms can be neglected. Meanwhile, lower order terms represent constants that introduce delay across the entire frequency range and thus can also be omitted. As a result β_2 (also known as group velocity dispersion), serves as a main culprit of pulse broadening and signal distortion in optical fibers and must be accounted for in our model.

After retaining only the β_2 term, the chromatic dispersion transfer function in the frequency domain is:

$$H_{CD}(\omega) = \exp\left(j \frac{\beta_2 \omega^2 L}{2}\right) \quad (2.7)$$

where L is the fiber length and ω represents the baseband frequency. This transfer function has several important properties:

- **Linear phase dependence on length:** The phase shift is proportional to fiber length L , meaning dispersion accumulates linearly with distance
- **Quadratic phase dependence on frequency:** Higher frequency components experience greater phase shifts, leading to differential group delays
- **Unity magnitude:** $|H_{CD}(\omega)| = 1$ for all frequencies, indicating that CD is a purely phase-distorting effect without attenuation, also known as an *allpass filter*.

The relationship between β_2 and the dispersion parameter D commonly used in fiber specifications is:

$$D = -\frac{2\pi c}{\lambda^2} \beta_2 \quad (2.8)$$

where c is the speed of light and λ is the operating wavelength. For standard single-mode fiber (SSMF) at 1550 nm, $D \approx 17$ ps/(nm·km). The accumulated dispersion over a link is:

$$D_{\text{total}} = D \cdot L \quad (2.9)$$

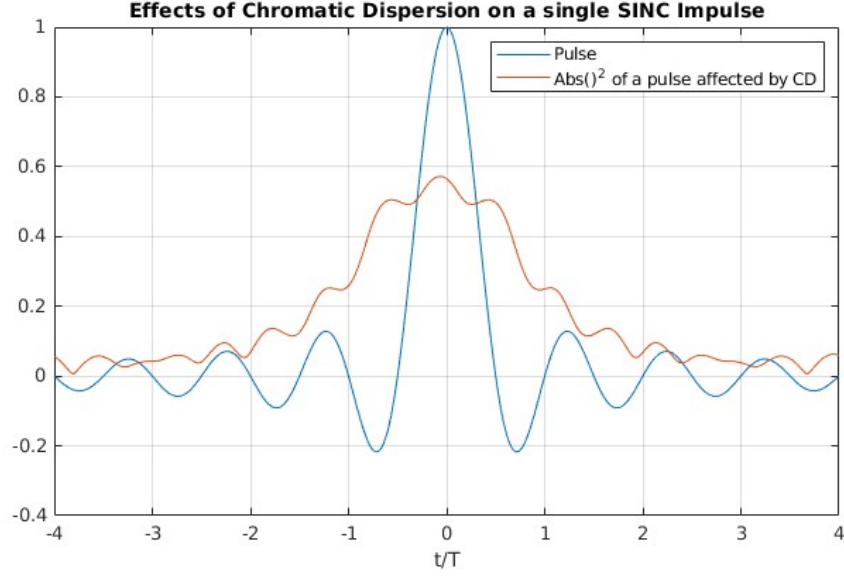


Figure 2.2: Pulse broadening and distortion caused by chromatic dispersion. As pulses propagate through the dispersive fiber, they spread temporally, leading to intersymbol interference when adjacent pulses overlap.

The primary consequence of chromatic dispersion (under the assumptions of the input signal being real-valued) is pulse broadening, where optical pulses that are initially narrow become temporally spread as they propagate. When adjacent pulses overlap, *intersymbol interference* (ISI) occurs, causing the receiver to incorrectly reconstruct the transmitted signal and leading to increased bit error rate (BER). The severity of this effect increases with both data rate (narrower pulses have broader spectra) and transmission distance (more accumulated dispersion), making CD the primary limitation for high-speed transmission beyond certain distances in IM/DD systems.

2.1.3 Signal definition

At this point, it is important to get a better understanding of the previously defined signal $x(t)$. We can make the assumption that the initial message consists of symbols for the alphabet $A = \{0, 1\}$, which are uniformly distributed. Initially, this allows us to encode it using a series of Dirac impulses. In this case, T is the *symbol rate*, describing the rate at which the symbols are sent through the channel.

$$s(t) = \sum_{i=0}^k \delta(t - i \cdot T) \cdot s_i, \quad s_i \in A \quad (2.10)$$

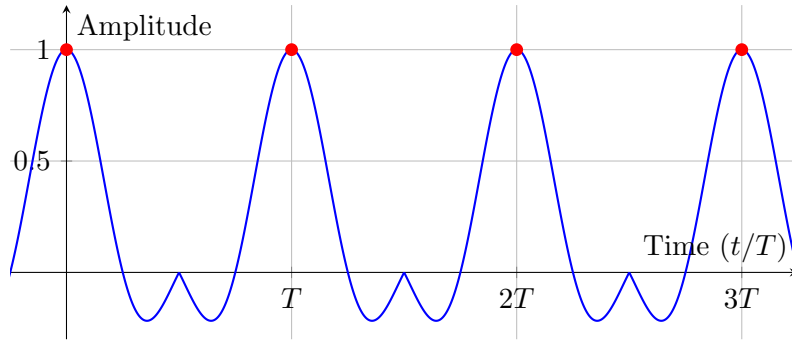


Figure 2.3: Result of a convolution of a sinc-type pulse shape with a series of Dirac impulses

The Dirac impulse is another essential component of communication systems, however it generally serves as a theoretical concept and emulating it in a real-life setting is generally not possible. Furthermore, it possesses some inconvenient mathematical properties, such as infinite-breadth spectrum, requiring a channel without any bandwidth limitations. This makes $s(t)$ unsuitable for transmission through the channel, thus it needs to be convolved with a pulse shaping filter $f(t)$ before being propagated. This is commonly referred to as *pulse shaping*. The resulting signal $x(t)$ can be defined as follows:

$$x(t) = f(t) * s(t) = \sum_{i=0}^k s_i \cdot f(t - i \cdot T), \quad s_i \in A \quad (2.11)$$

By definition, convolving a signal with a series of Dirac impulses produces a sequence of shifted replicas of that signal. Pulse shaping is an area of ongoing research in signal processing and communication systems, as various shapes may provide different advantages due to their frequency and time-domain response, susceptibility to noise, etc.

In order to encode the message, we need to be able to represent different values with each pulse. This requires us to multiply the resulting pulses with different values. This process is commonly referred to as *modulation* and is another area that has undergone rigorous research and optimization. Due to the nature of our system (see 2.1.5), we only have to deal with pulses that represent either a 1 or a 0 and multiply them with the corresponding value. This represents an extremely basic variant of *amplitude modulation* called *On-Off-Keying* (OOK), meaning that the amplitude of a pulse is the differentiating component between the symbols that it carries. This approach is very simple to implement and is robust against distortion, as the distance between different symbols is relatively high. However, its obvious downside is low data throughput, as only a single bit is encoded per pulse. Some of the common approaches in optical communication in-

clude *pulse amplitude modulation* (PAM) (see 2.4) and *quadrature amplitude modulation* (QAM). Different modulation methods attempt to balance out the the signal's data-rate and its susceptibility to noise by utilizing more than 2 levels of amplitude modulation in a single pulse, or encoding additional information into the symbol's phase.

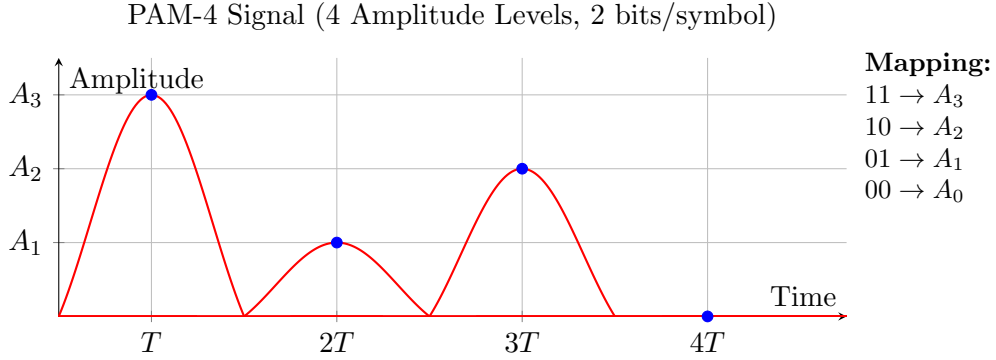


Figure 2.4: Example of amplitude modulation (PAM-4) being utilized to encode 2 bits of information in a single pulse.

Reconstruction and Detection

Reconstruction and detection is a task that is performed at the systems receiver and is essential for the user to be able to decode the message that has been sent. After passing through the receiver, the signal must be processed to recover the original data sequence. This process consists of two basic steps: *sampling* and *decision*.

Let us define the received signal as $r(t)$. At the point of its arrival, this signal will be affected by various disruptive effects. $r(t)$ is sampled at the sampling intervals $t = kT$, where T represents the timing offset initially used by us to construct the series of Dirac impulses $s(t)$. Ideally, these sampling instants correspond to the centers of the received pulses, where the signal-to-noise ratio is maximized. Let r_k denote the sample at time k :

$$r_k = r(kT) \quad (2.12)$$

The decision process then maps each sample r_k to the nearest valid symbol through threshold detection. For OOK modulation, this involves comparing the sample against a decision threshold t . In this case, an assumption about the probability of receiving 1 and 0 being equal can be made. Thus we declare $t = 0.5$ and define the obtained value at point k to be a_k

$$a_k = \begin{cases} 1 & \text{if } r_k \geq 0.5 \\ 0 & \text{if } r_k < 0.5 \end{cases} \quad (2.13)$$

where $\hat{a}_k$ is the estimated data symbol. In practice, various estimation techniques can be used to obtain an optimal threshold in order to account for channel properties, as well as the characteristics of the message itself.

2.1.4 Noise

The performance of a communication system is fundamentally limited by various disruptive effects. One of the most common types is additive noise. It is typically modeled as *additive white Gaussian noise* (AWGN)—a stochastic process with an underlying Gaussian distribution, with its spectrum containing an infinite range of frequencies. In optical communication systems, noise frequently manifests itself as thermal noise in the receiver electronics, which affects the electrical signal after photodetection (see Section 2.1.5). Other types of noise commonly present in optical systems include signal-dependent noise and optical noise. However, due to the channel length modeled within our implementation (see 3), both of these can be neglected.

In our model, we define the random noise as $n(t)$. Adding it to the signal $x(t)$ yields $x_n(t)$ —a signal distorted by Gaussian noise:

$$x_n(t) = x(t) + n(t) \quad (2.14)$$

$$n(t) \sim \mathcal{N}(0, \sigma^2) \quad (2.15)$$

where $\mathcal{N}(0, \sigma^2)$ denotes a Gaussian distribution with zero mean and variance σ^2 .

Noise Power and Variance

The noise power P_n is an important parameter in signal processing that is directly related to the variance σ^2 of the underlying Gaussian distribution. Together with the signal power S_n , it lets us assess to which extent the information carried by the signal is suppressed by the noise (see 2.1.4). For a zero-mean noise process $n(t) \sim \mathcal{N}(0, \sigma^2)$, the average power is equal to the variance:

$$P_n = \mathbb{E}[n^2(t)] = \sigma^2 \quad (2.16)$$

where $\mathbb{E}[\cdot]$ denotes the expected value. Higher variance corresponds to stronger noise,

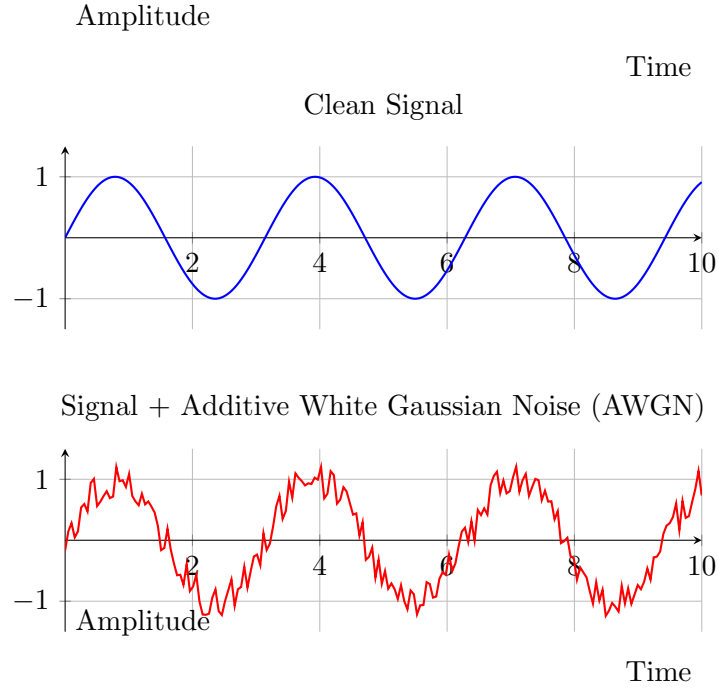


Figure 2.5: Effect of additive white Gaussian noise on a clean sinusoidal signal. The noisy signal exhibits random fluctuations around the original waveform.

which degrades the signal quality and increases detection errors. Adjusting σ^2 , will allow us to simulate different noise conditions and evaluate system performance under varying levels of noise.

Signal-to-Noise Ratio and Bit Error Rate

The *signal-to-noise ratio* (SNR) quantifies the relative strength of the signal compared to the noise and is a key metric for characterizing system performance. SNR is typically expressed in decibels (dB):

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \left(\frac{P_s}{P_n} \right) = 10 \log_{10} \left(\frac{P_s}{\sigma^2} \right) \quad (2.17)$$

where P_s is the average transmitted signal power. Higher SNR indicates better signal quality, as the signal is stronger relative to the noise. The *bit error rate* (BER) is a metric that quantifies the probability that a transmitted bit is incorrectly detected at the receiver. BER is fundamentally determined by the SNR: higher SNR leads to lower BER, as the noise is less likely to cause the received sample to cross the decision threshold and result in a wrong bit decision.

Real-world receiver implementations will typically attempt to reduce the noise and improve signal quality before sampling and detection. This can be done in a variety of ways but will generally involve restricting the bandwidth of the channel, such that the frequencies that lie outside the range of the original signal are attenuated. This is commonly referred to as *filtering*. This approach can be extremely effective but will never remove the entirety of the white Gaussian noise, as its definition implies that it is distributed uniformly across all frequencies, including the ones that belong to the signal itself.

A *lowpass filter* is commonly employed to pass signal frequencies while attenuating high-frequency noise components. The ideal lowpass filter has a rectangular frequency response:

$$H_{\text{LPF}}(f) = \begin{cases} 1 & |f| \leq f_c \\ 0 & |f| > f_c \end{cases} \quad (2.18)$$

where f_c is the cutoff frequency, typically chosen to encompass the signal bandwidth while rejecting out-of-band noise.

The effectiveness of the lowpass filter can be understood through the concept of *power spectral density* (PSD). For a wide-sense stationary signal $x(t)$ with average power:

$$P_s = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |x(t)|^2 dt \quad (2.19)$$

the PSD $S_x(f)$ describes how this power is distributed across frequencies:

$$S_x(f) = \lim_{T \rightarrow \infty} \frac{1}{T} |X_T(f)|^2 \quad (2.20)$$

where $X_T(f) = \mathcal{F}\{x_T(t)\}$ is the Fourier transform of the truncated signal over the interval $[-T/2, T/2]$. The total signal power is then recovered by integrating the PSD over all frequencies:

$$P_s = \int_{-\infty}^{\infty} S_x(f) df \quad (2.21)$$

One of the possible filtering approaches involves implementing a frequency cutoff f_c , that allows us to retain a specified percentage of the total signal power. Removing frequency components beyond f_c will reduce the total noise power while preserving most of the signal power. For white noise with power spectral density $N_0/2$, the filtered noise power for a double-sided spectrum is:

$$P_{n,\text{filtered}} = N_0 \cdot f_c \quad (2.22)$$

Thus, reducing the filter bandwidth directly reduces the noise power, improving the SNR and consequently the BER. However, the filter must be carefully designed on a case-by-case basis after having studied the spectral properties of the original signal. If f_c is too small, significant signal energy is lost, causing signal distortion and potentially introducing even more errors.

It is worth noting that while this approach can effectively mitigate additive noise, it does not address other types of distortions that occur within an optical channel (see 2.1.2) and introduce an *noise floor* that cannot be addressed by simply restricting specific frequencies.

2.1.5 IM/DD Systems

Intensity modulation/direct detection (IM/DD) is widely used type of optical communication systems. particularly for short-reach networks. As introduced in Chapter 1, IM/DD systems are used for their simplicity and cost-effectiveness compared to more complex alternatives such as *coherent detection systems*, which detect both intensity and phase but come with an increased complexity when implemented in real world systems.

In an IM/DD system, information is encoded solely in the optical intensity at the transmitter. At the receiver, a *photodiode* performs direct detection, converting the optical signal into electrical current. This is known as square-law detection, where the generated photocurrent $i(t)$ is proportional to the optical power:

$$i(t) \propto |E(t)|^2 \quad (2.23)$$

where $E(t)$ is the complex envelope of the optical field. This square-law detection has a fundamental consequence: *all phase information is lost*. If the optical field is represented as $E(t) = A(t)e^{j\phi(t)}$, the detected signal becomes:

$$i(t) \propto |A(t)e^{j\phi(t)}|^2 = A^2(t) \quad (2.24)$$

The phase term $e^{j\phi(t)}$ disappears, leaving only the intensity information $A^2(t)$. The inability to extract the phase outside of additional signal processing limits IM/DD systems to intensity-based modulation formats. IM/DD systems typically utilize OOK modulation due to its simplicity and alignment with the intensity-only constraint. Advanced modulation formats like them (QAM) become incompatible with direct detection for this reason.

The communication systems, the *baud rate* refers to the number of symbols transmitted per second, while the *bit rate* refers to the number of information bits per second. In

a typical IM/DD system utilizing OOK modulation, a single bit is transferred with each pulse (symbol). As a result, these quantities will be perceived as equal. For example, a 10 Gbaud OOK system transmits 10 Gbps of data.

2.2 Gerchberg-Saxton Algorithm

The general Gerchberg-Saxton (GS) algorithm was initially introduced in 1972 as an iterative method for retrieving unknown phase information from intensity measurements [GS72]. It exploits the fact that intensity measurements are available in both the spatial domain, as well as its Fourier transform. The GS algorithm alternates between these domains using FT and IFT, continuously enforcing known amplitude constraints in each domain while preserving the initially generated phase (see 2.2). Through this process, the algorithm converges towards an approximation of the original phase. The convergence is often inconsistent due to the initial phase being randomly generated, as well as the algorithm having multiple local minima. As demonstrated in this thesis, the GS algorithm can be repurposed for chromatic dispersion compensation through predistortion.

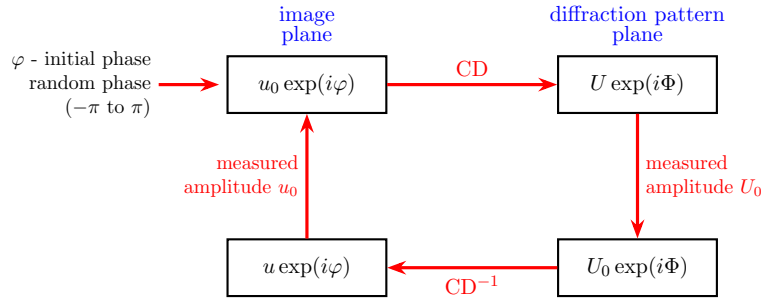


Figure 2.6: Sketch showcasing the principle behind the GS algorithm.

2.3 Repurposed GS Implementation

A repurposed implementation of the GS Algorithm is proposed in a paper by Zou et.al. [ZLW⁺22]. Instead of estimating the missing phase, the initially unknown amount of predistortion is estimated by continuously iterating and enforcing a constraint, which has also been adjusted to reflect the nature of underlying system. Finally, instead of using the spatial and Fourier domains, the modified algorithm alternates between the transmitter plane (before dispersion) and receiver plane (after dispersion) by applying the chromatic dispersion transfer function and its inverse. We will define them as functions from now

on, with $X(f)$ being the signal on either end of the receiver.

$$\text{CD}(X(f)) = \exp\left(j\frac{\beta_2\omega^2 L}{2}\right) \cdot X(f) \quad (2.25)$$

$$\text{ICD}(X(f)) = \exp\left(-j\frac{\beta_2\omega^2 L}{2}\right) \cdot X(f) \quad (2.26)$$

The constraints used in the standard GS algorithm can also be readjusted to better fit our case. First constraint can be derived from the fact that the input signal of the system is inherently real-valued. We can make use of that in order to enforce a zero phase on the transmitter side during each loop iteration. Additionally, any phase information is destroyed by the photodiode's square-law detection: $i(t) \propto |E(t)|^2$, as soon as the signal leaves the channel. This can be used to enforce a target amplitude with disregard to phase, similar to the original GS algorithm. These constraints also allow for this algorithm to be realized via different approaches, further explained down below.

2.3.1 Amplitude-only predistortion

The amplitude-only predistortion mimics the approach described in the [ZLW⁺22]. In this case we make an assumption about the incoming pulse not having any additional phase. As such, the constraint at the transmitter resets the phase to 0 every iteration. At the receiver, we enforce the desired pulse amplitude with an arbitrary phase. We assume that the phase is of little relevance, as it is removed entirely in the next step via square-law detection. The presented approaches are very similar but may have varying efficacy in different scenarios.

Algorithm 1 Modified GS Algorithm with Amplitude-Only Predistortion

Require: Target amplitude $A_t(t)$, maximum iterations $N_{\max}$

Ensure: Predistorted transmitter signal $s_{\text{tx}}(t)$

- 1: Initialize: draw random phase $\varphi_0(t) \sim \mathcal{U}(0, 2\pi)$
 - 2: $s_{\text{rx}}(t) \leftarrow A_t(t) \cdot \exp(j\varphi_0(t))$
 - 3: **while** error criterion is not satisfied **do**
 - 4: $s_{\text{tx}}(t) \leftarrow \text{ICD}\{s_{\text{rx}}(t)\}$
 - 5: $s_{\text{tx}}(t) \leftarrow |s_{\text{tx}}(t)| \cdot \exp(j \cdot 0)$
 - 6: $s_{\text{rx}}(t) \leftarrow \text{CD}\{s_{\text{tx}}(t)\}$
 - 7: $s_{\text{rx}}(t) \leftarrow A_t(t) \cdot \exp(j \arg[s_{\text{rx}}(t)])$
 - 8: **end while**
 - 9: **return** $s_{\text{tx}}(t)$
-

2.3.2 Phase-Only predistortion

Phase-only predistortion approach builds off of the logic of the GS algorithm in a slightly different way. The enforced constraint on both sides of the channel is the amplitude, while the phase is reutilized from the previous transformation. Upon finishing, this algorithm produces a signal with an enforced amplitude and a phase, which should result in the signal being chromatically dispersed in a "correct" way.

Algorithm 2 Modified GS Algorithm with Phase-Only Predistortion

Require: Target amplitude $A_t(t)$, maximum iterations $N_{\max}$

Ensure: Predistorted transmitter signal $s_{\text{tx}}(t)$

- 1: Initialize: draw random phase $\varphi_0(t) \sim \mathcal{U}(0, 2\pi)$
 - 2: $s_{\text{rx}}(t) \leftarrow A_t(t) \cdot \exp(j\varphi_0(t))$
 - 3: **while** error criterion is not satisfied **do**
 - 4: $s_{\text{rx}}(t) \leftarrow \text{CD}\{s_{\text{launch}}(t)\}$
 - 5: $s_{\text{rx}}(t) \leftarrow A_t(t) \cdot \exp(j \arg[s_{\text{rx}}(t)])$
 - 6: $s_{\text{tx}}(t) \leftarrow \text{ICD}\{s_{\text{rx}}(t)\}$
 - 7: $s_p(t) \leftarrow A_t(t) \cdot \exp(j \arg[s_{\text{tx}}(t)])$
 - 8: **end while**
 - 9: **return** $s_{\text{launch}}(t)$
-

2.3.3 Convergence Properties

The convergence behavior depends on the predistortion strategy and initialization conditions. For our purposes, two convergence criteria can be utilized:

Fixed iteration count: The algorithm runs for a predetermined $N_{\max}$ iterations, ensuring consistent computational cost but potentially terminating too early or past plateauing.

RMSE threshold: The algorithm monitors the root mean squared error between consecutive iterations or between the current and target signal:

$$\text{RMSE}^{(k)} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(x_i^{(k)} - x_{i,\text{target}} \right)^2} \quad (2.27)$$

and terminates when $\text{RMSE}^{(k)} < \epsilon$, where ϵ is a predefined threshold. If the desired RMSE threshold is not reached within $N_{\max}$ iterations, the algorithm terminates at the maximum iteration count, returning the best solution found.

Convergence may end up being inconsistent between runs due to random initialization of the phase $\varphi_0(t) \in [-\pi, \pi]$. Different initial phases lead to different local minima, potentially causing significant variation in both convergence speed and final RMSE. We

expect the algorithm to initially exhibit rapid initial descent (5-10 iterations), then gradual refinement, and to eventual plateau at a local minimum.

Due to this variability, it can be assumed that algorithm does not consistently reach the global optimum. The convergence characteristics of this algorithm are compared and evaluated in the corresponding chapter 4.

3 Implementation

This chapter describes the implemented components within the simulation, along with the testing methodology used. We put the simulated system in real-world context, in an attempt to highlight limitations that occur when a physical system is being modeled. Additionally, similar approaches from existing literature are referenced in order to aid us with our goal implementation.

3.1 Optical channel simulation

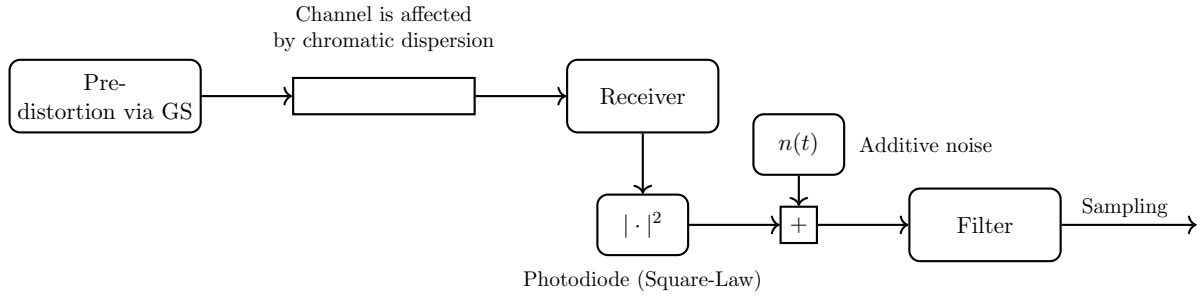


Figure 3.1: Simplified model of the simulated system

Our implementation follows the model of an optical IM/DD Channel defined in 2.1. We implement all necessary components as blocks that propagate the simulated signal. The main goal was to highlight the effects of using predistortion to mitigate CD under conditions that closely approximate those of the real-world.

1. Apply pulse shaping filter to generate transmitted signal $x(t)$
2. Propagate through dispersive channel (apply $H_{CD}(\omega)$)
3. Perform square-law detection at photodiode: $i(t) \propto |x(t)|^2$
4. Add thermal noise: $r(t) = i(t) + n(t)$
5. Apply lowpass filter to remove high-frequency noise

6. Sample at instants $t = kT + t_0$
7. Apply threshold detection to recover transmitted symbols

Simulating continuous-time physics systems in discrete digital environments requires careful attention to the sampling rate.

In accordance to the Nyquist-Shannon sampling theorem, a signal with bandwidth B must be sampled at $f_s \geq 2B$ (see 3.2) to avoid aliasing. In simulation, we employ *oversampling* with an oversampling factor $N_{os} = f_s/R_b$, where R_b is the bitrate of the system. Since we are working with an OOK system, where the Baud rate equals the bitrate, R_b can be replaced with the symbol rate R_{sym} . This provides sufficient resolution to accurately model the effects of noise and CD, as well as represent pulse shapes even at very high data rates commonly present in optical communication systems.

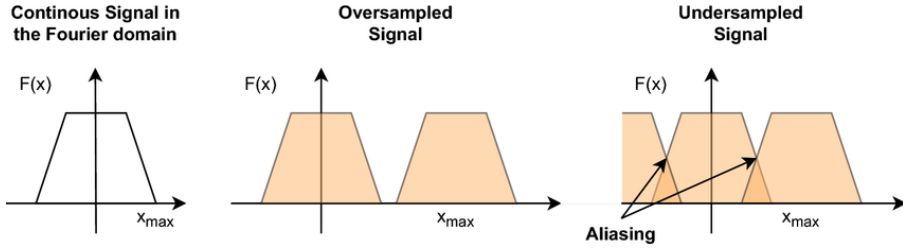


Figure 3.2: An example of insufficient sampling rate causing aliasing [GKK22].

Furthermore, a lot of operations that would normally occur in a continuous time-domain have to be performed using their discrete-time version. As an example, discrete-time convolution is defined as follows:

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot h[n - k] \quad (3.1)$$

The simulation described in this paper has been implemented and tested in Matlab, which provides a rich interface of functions such as noise generation, convolution and integration. Some of references to the implementation are listed at the end of this section.

Pulse Shaping

We implement seven pulse shapes, each representing different trade-offs between time-domain compactness and frequency-domain bandwidth:

- Rectangular (RECT)

- Raised cosine (RCOS)
- Root-raised cosine (RRCOS)
- Sinc (SINC)
- Cosine-squared (COS_SQR)
- Gaussian (GAUS)

Rectangular and sinc pulses represent extreme cases. The rectangular pulse is maximally compact in time but has an infinitely broad spectrum, making it sensitive to bandwidth limitations and phase distortion via CD. The sinc pulse is perfectly bandlimited in frequency but extends infinitely in time. Practical pulse shapes, that are applied in real-world implementations, offer controlled bandwidth with minimal ISI.

Testing across this range of pulse shapes allows evaluation of the GS algorithm's robustness to different spectral properties and identifies which characteristics most influence dispersion compensation performance.

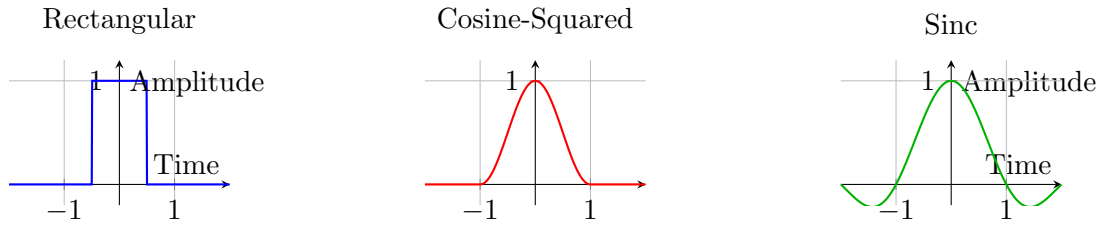


Figure 3.3: Three representative pulse shapes showing the spectrum-duration trade-off: rectangular (broad spectrum, compact time), cosine-squared (balanced), and sinc (narrow spectrum, extended time).

3.1.1 Filter Implementation

The filter is implemented via simple integration over the initially sent signal PSD. In order to keep it consistent, we calculate the corresponding area in percent prior to the application of any effects like noise or CD. The resulting filter has varying levels of efficacy at different percentage levels for different pulse shapes. This is due to the fact that pulse shapes with a very wide PSD capture a lot of noise across all frequencies. Counterintuitively, using a very high filter percentage for would result in a lot of excessive noise in such case.

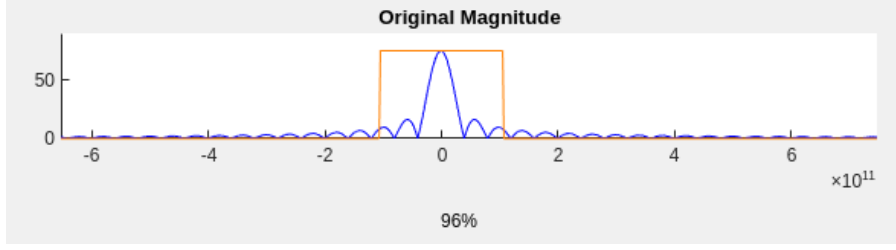


Figure 3.4: Perfect lowpass capturing 96% of the signal's PSD

We include a number of other variables that can be adjusted during tests, while attempting to stick to default values that could be representative of the system's real world implementation. We model a 30 Gbps IM/DD system with symbol period $T = 30$ ps (one bit per symbol in OOK). The initial sampling frequency of $f_s = 3$ THz provided an oversampling factor N_{os} of 100 samples per symbol (at the rate of 30Gbps). A high N_{os} provides a good resolution but is generally computationally expensive for operations like convolution and multiplication. The fiber length is $L = 20$ km, typical for metropolitan networks where chromatic dispersion becomes significant at high data rates. The carrier wavelength $\lambda = 1550$ nm corresponds to the C-band used in optical communications; in our simulation, it appears only as a parameter in the dispersion transfer function $H_{CD}(\omega)$, as we model the baseband envelope signal directly. The test sequence contains 400,000 symbols providing sufficient length for statistical analysis. The dispersion severity depends on the product of data rate squared and fiber length: B^2L .

Parameter	Default Value
Symbol rate R_{sym}	30.3 MBaud
Oversampling Rate N_{os}	100x
Fiber length L	20 km
Carrier wavelength λ	1550 nm
Speed of light c	3×10^8 m/s
Test signal length N	400,000 symbols

Table 3.1: Simulation parameters for IM/DD system with CD.

4 Evaluation

Evaluation details

In this section we evaluate the two different implementations of the modified GS algorithm under various conditions. In order to assess its efficacy we establish a set of default values, similar to ones that could occur in a real-world system. We then proceed to evaluate the impact of CD and GS predistortion on the accuracy of transmission by using plots, as well as numerical estimations.

4.1 Exploring Baseline Performance

Before evaluating the effectiveness of the modified Gerchberg-Saxton algorithm for chromatic dispersion compensation, it is essential to establish the baseline performance characteristics of the uncompensated IM/DD system. This baseline provides a reference against which the benefits of predistortion can be quantified and demonstrates the severity of chromatic dispersion at the chosen operating parameters.

The baseline system operates with the default parameters specified in Table 3.1.1. No dispersion compensation is applied—the transmitted signal consists of shaped pulses without predistortion. The system is evaluated across a range of signal-to-noise ratios (SNR) from -10 to 18 dB to characterize performance under varying noise conditions. This is important to establish the baseline functionality of our simulation and check that it behaves in accordance with the expectations. The results can be seen in 4.1

Additionally we evaluate our lowpass filter performance on a single pulse shape (Sinc), in order to confirm that increasing the percentage of the PSD that is being covered improves BER (see 4.2).

4.2 Effects of Chromatic Dispersion

Chromatic dispersion introduces deterministic pulse broadening that fundamentally differs from random additive noise. As pulses propagate through the channel, they spread temporally, causing adjacent symbols to overlap and create ISI. Unlike noise, which can

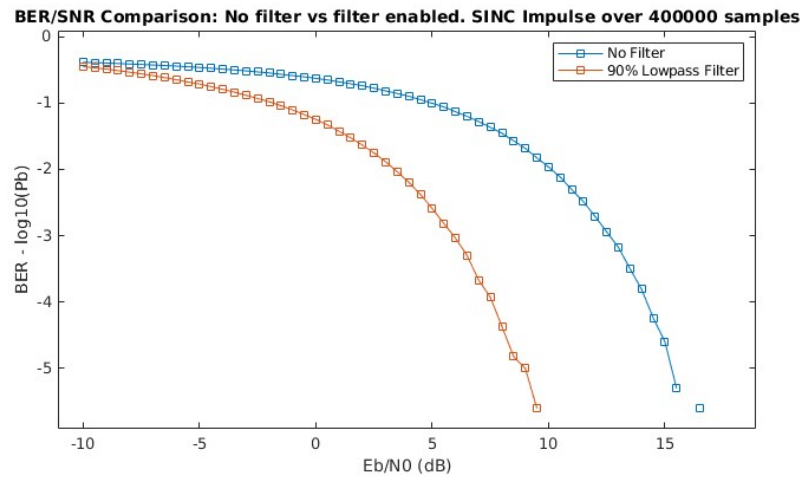


Figure 4.1: Evaluating the efficacy of a lowpass filter on a basic system configuration

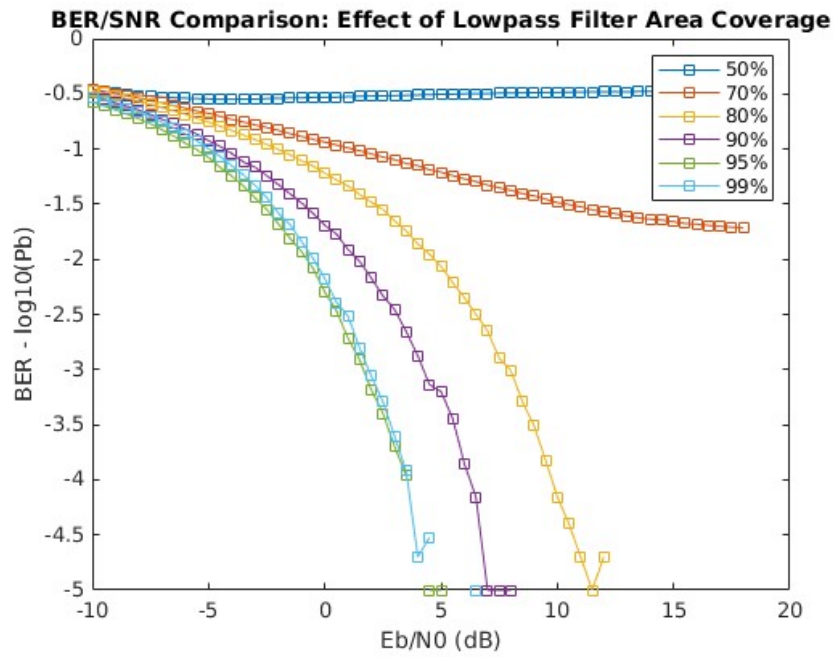


Figure 4.2: Comparing different percentages used for the lowpass filter

be partially filtered, this temporal spreading distorts the signal structure itself within the signal bandwidth.

The lowpass filter removes high-frequency noise to improve SNR, but it cannot reverse chromatic dispersion. The pulse overlap caused by dispersion exists within the pass-band, not in the out-of-band noise spectrum, making frequency-domain filtering ineffective against ISI. Even in a noiseless channel, overlapping pulses prevent correct symbol detection at the sampling instants. This behaviour can be clearly seen in, as the values within the BER plot flatline past a certain point, regardless of the SNR. This manifests in BER curves as an error floor.

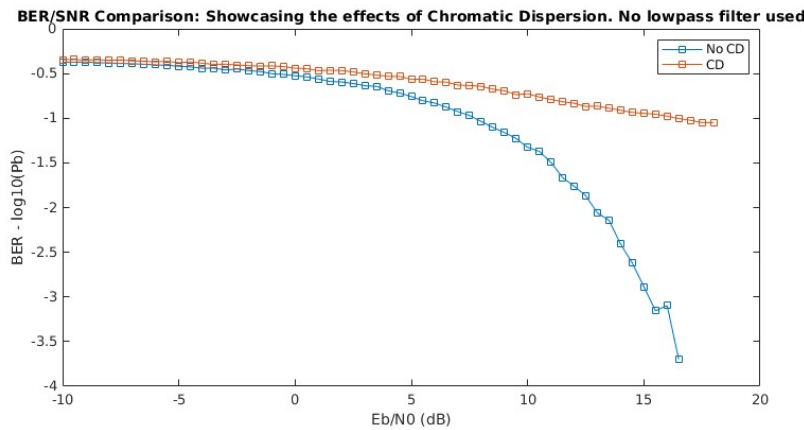


Figure 4.3: An example of a BER curve flatlining due to ISI caused by CD

4.3 Exploring the GS Algorithm

In order to assess the efficacy of our algorithm, we used both BER plots and scatter plots. Scatter plots provide a nice overview of how close the points group towards the actual possible values (1 and 0). As shown in 4.4 and 4.5, the phase only predistortion seems more promising, as it forms tighter clusters over the intended values.

The final tests were carried out using the amplitude-only predistortion approach (see 4.6, 4.7, 4.8), which was the one we initially copied from the referenced paper [ZLW⁺22]. Here we see promising results with the default system configuration. Interestingly enough, the application of the algorithm sets an arbitrary "baseline", as the predistorted BER curve looks similar in all plots, regardless of the channel length. This approach seems to make rather small difference in shorter channel lengths (see ??). However for longer channel lengths, we see a noticable improvement over what would otherwise be a heavily distorted signal.

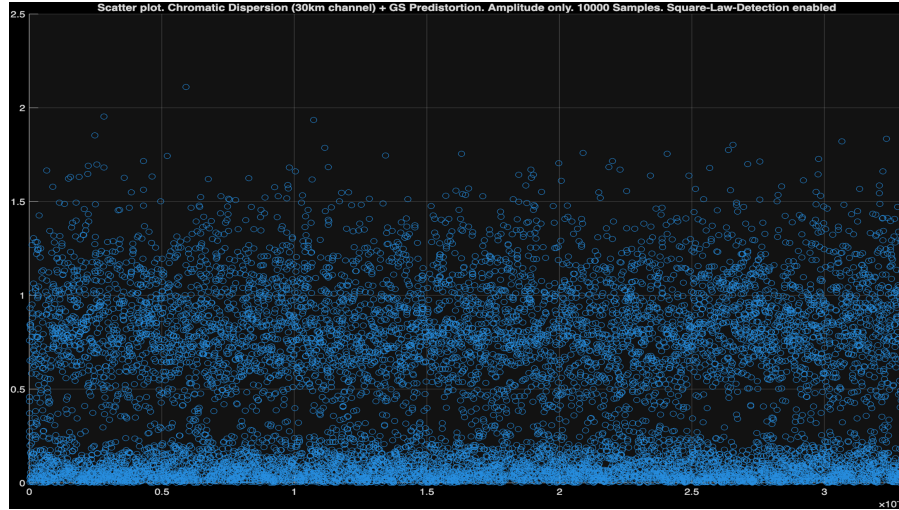


Figure 4.4: Scatterplot for a 10000 sample test using the amplitude-only predistortion

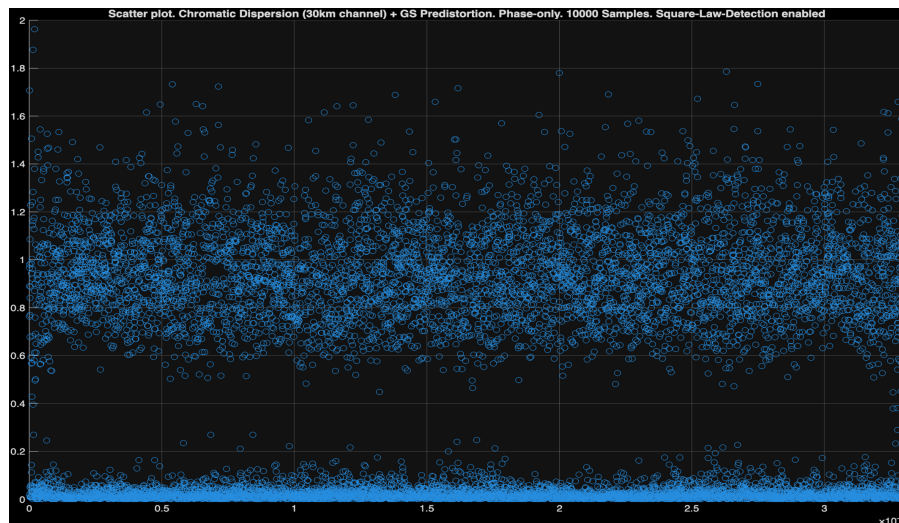


Figure 4.5: Scatterplot for a 10000 sample test using the phase-only predistortion. Notice the improved performance over the amplitude-only approach.

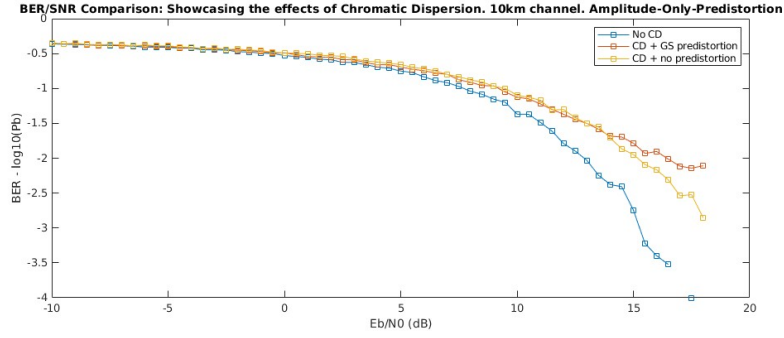


Figure 4.6: CD compensation over a 10km channel. Amplitude-only predistortion. Notice the minimal difference between the signal with and without predistortion

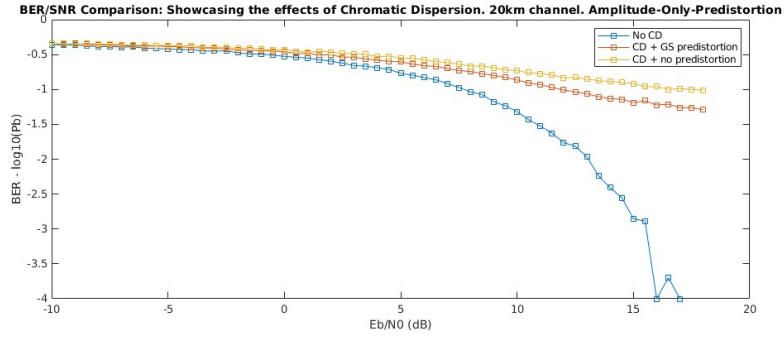


Figure 4.7: CD compensation over a 20km channel. Amplitude-only predistortion.

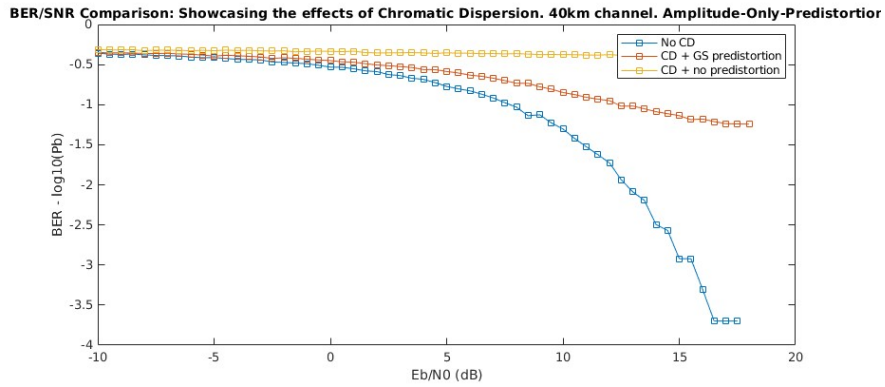


Figure 4.8: CD compensation over a 40km channel. Amplitude-only predistortion.

5 Conclusions

The following thesis introduces the application of the GS algorithm to the problem of CD compensation via predistortion. We introduce two different ways to implement the algorithm and analyze its performance via plots.

Based on the results obtained in our tests, it is safe to say that this approach is able to eliminate CD to some degree and can serve as a helpful and simple-to-implement alternative or supplement to other solution used for signal predistortion.

We came upon some unexpected results when testing different approaches with various configurations. Obviously, this work presents a very large number of parameters that can be tested, ranging from various pulse shapes to channel lengths. In best case scenarios, we were met with adequate performance.

In conclusion, the approach of predistorting the signal via the modified Gerchberg-Saxton Algorithm in order to mitigate chromatic dispersion shows some promise and offers a very simple implementation. Further research is needed to establish the best possible environment for this approach to be used in. It is possible that in the right conditions. The potential for improvement goes far beyond the simple overview provided within this thesis.

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